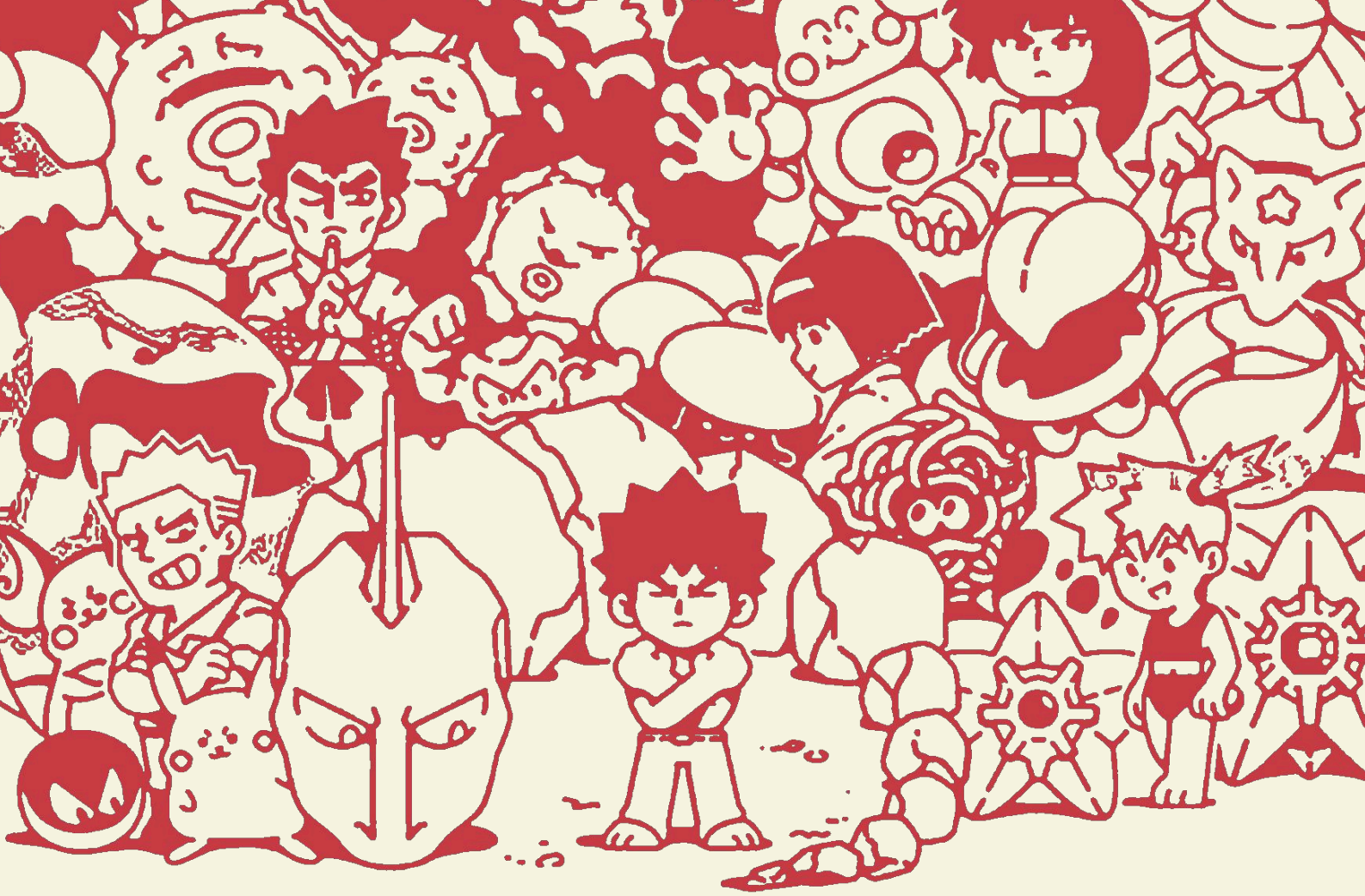
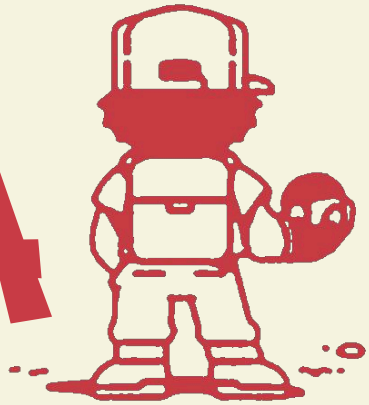




GOTTA CODE EM ALL

ODT CAPSTONE PROJECT

Reflections

Sachit Patnaik
Tanay Dhavate

BITS Design
2025-29

TEAM REFLECTION

How did the team work together?

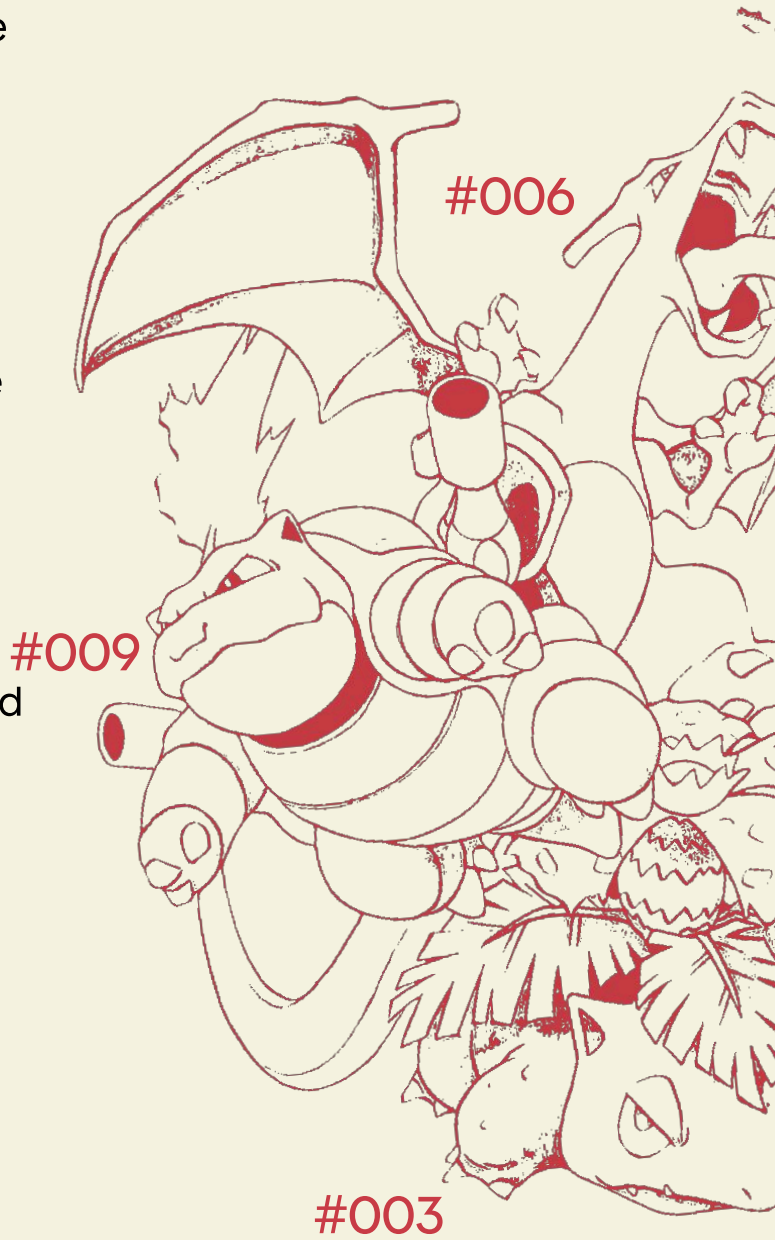
- The team worked well together with very few conflicts , and those that did arise were only over trivial matters like presentation style, nothing substantial
- Our shared interest in Pokémon and interactive electronics meant we were naturally aligned on the direction of the project from the start
- Tasks divided cleanly between hardware and software, with both members contributing across both where needed

What worked well?

- Shared enthusiasm for the subject made collaboration feel natural rather than forced
- Decision-making was fast because we agreed on the core vision early and rarely second-guessed it
- The division between physical build and software work played to each person's strengths

What could be improved?

- Very little any improvements would be minor and none relate to team dynamics
- Could have documented decisions and testing results more consistently throughout the process



SACHITS REFLECTION

What I Learned

• How to work through genuinely complex embedded systems problems independently , reading datasheets, testing hypotheses, and debugging without a clear answer in sight

• That hardware and software failures are rarely obvious; the real skill is staying methodical when everything seems broken

• How to integrate multiple protocols I2C, SPI, I2S, ADC, GPIO on a single microcontroller without conflicts

What I Struggled With

• The last five days of the project were almost entirely spent troubleshooting the codebase , getting all systems to run together stably was far harder than getting any one of them working alone

• ODT as a module can be uniquely frustrating in a way that other modules are not it is not about effort or attitude, but about systems being genuinely complex in ways that are hard to even diagnose, let alone fix

• There were long stretches where I had no idea what was going wrong, which tested patience more than any other module has

What I Am Proud Of

• When the entire system finally came together and ran end to end for the first time, it was one of the most rewarding moments I have had in this course genuinely emotional after everything it took to get there

• Proud of the fact that we did not take shortcuts, every component, every animation, every audio file was figured out properly

• Overall this was a 10/10 experience and one of my favourite modules, the frustration was real but so was the satisfaction on the other side of it

#724

TECHINICHAL REFLECTION

Key Technical Challenges

• Getting the I2S amplifier to output clean audio was the most difficult hardware problem , distortion persisted across multiple file formats and settings

• Converting GIFs into bitmap sequences and then into raw binary frame files required a custom pipeline and manual adjustment of light/dark inversion per Pokémon

• Running out of GPIO pins forced a workaround using pins 34 and 35, which are input-only with no internal pull-up resistors

What Did We Learn?

• Learned to work through component documentation and data sheets independently, figuring out the OLED and I2S amplifier without handholding was challenging but deeply rewarding

• Understood how non-blocking timing works in practice .replacing sleep() with ticks_diff() to allow multiple outputs to run simultaneously

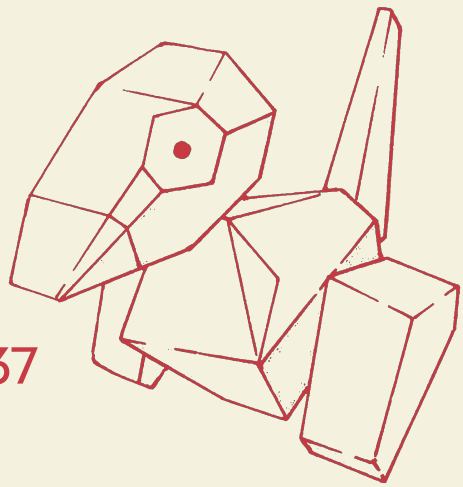
• Gained hands-on experience with ADC stability, I2S audio streaming, binary frame rendering, and SD card file management on a microcontroller

Mistakes & Debugging

• GPIO 12 on the ESP32 nearly damaged the board ,I earned to check pin boot modes before using any GPIO for peripherals

• Spent significant time debugging audio distortion before discovering the fix: a 75Ω resistor between the GPIO and the data pin of the amplifier, which resolved the issue immediately

• Multiple crash cycles taught us to always initialise peripherals sequentially with deliberate delays rather than all at once



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DESIGN REFLECTION

Did the experience match your intention?

• Largely yes ,the core fantasy of scanning a figure and catching a Pokémon came through clearly and users responded strongly to it
• The physical Pokéball shaking was the standout moment and landed exactly as intended

What Would You Improve?

• The placement and integration of the Pokéball into the overall form needs more refinement, it currently feels like an accessory rather than a natural part of the device
• With more time and another iteration of prints, the enclosure and physical layout would be significantly more considered
• The product would benefit from being treated as a fully resolved object , cleaner wiring, a custom PCB, and a more refined finish

User Feedback Insights

• Users found the interaction intuitive and engaging , most completed it unaided
• Repeat interactions were common, driven by the variety of Pokémon and the rarity system
• Jury feedback highlighted that more user research and ergonomic consideration would strengthen the design , particularly around how the user holds the device while also interacting with the Pokéball
• More rounds of user testing would inform better decisions around button placement, screen legibility at different angles, and the physical form of the figures

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